Ex.No:9 Date: 20/08/2026

Implementation of SUBNETTING in CISCO Packet tracer simulator.

Aim:

To implement subnetting in cisco packet tracer, design multiple subnet with a suitable number of hosts, assign static with a suitable numbers IP addresses & establish communication between different subnets using a router.

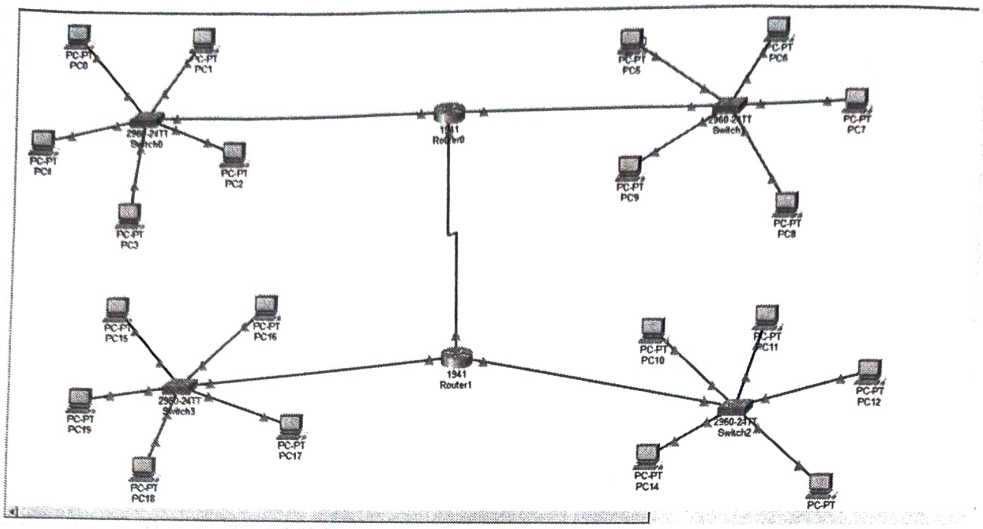
Introduction:

Subnetting is the process of dividing a larger network into smaller logical networks called subnets. Each subnet can contain a suitable number of hosts. Subnetting helps in organizing networks and allows different groups of hosts to communicate through a router.

In this experiment, multiple subnets are designed using cisco Packet Tracer. Static IP addresses are assigned to the hosts & router interfaces. The different subnets are then connected through a router & packet transmission is observed.

Requirements:

* Cisco Packet Tracer
* PCs / Hosts


PC-PT
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
1941
Router0
Switch1
Switch0
1941
Router1
PC10
PC11
PC12
PC13
PC14
Switch2
PC15
PC16
PC17
PC18
PC19
Switch3

DTP

* Router
* Network connections
* Suitable IP addresses.

Procedure :

Step 1 : Create the Network.

1. Open Cisco Packet Tracer.
2. Place the required number of PCs/hosts in the workspace.
3. Place a router to connect the different subnets.
4. Divide the hosts into multiple groups based on the required subnet design.
5. Connect the host belonging to each subnet.

Step 2 : Design Multiple Subnets

1. Create multiple subnets according to the required number of hosts.
2. Select a suitable IP network for the subnet
3. Divide the network into smaller subnetworks
4. Ensure that each subnet has sufficient IP addresses for its hosts.
5. Identify the network address and suitable host addresses for each subnet.

* Step 3 : connect the hosts belonging to each subnet using the required network connections.

* step 4: Assign suitable static IP addresses and subnet masks to all the hosts.

*. step 5: configure the appropriate default gatway for each host.

* step 6: connect the different subnets using a router.

*. step 7: configure the router interfaces with suitable IP addresses & subnet masks.

*. step 8: check the IP configuration and connectivity of all the devices.

*. step 9: perform packet transmission between hosts in the same subnet and different subnets.

*. step 10: use simulation mode to observe the packet travelling from one subnet to another through the router.

* step 11: Test packet communication when the subnets are connected through the router.

* step 12: Test communication without connecting the different subnets through a router and compare the results.

*. step 13: Verify that the router is required for communication between different subnets.

Result:

Thus, multiple subnets were successfully designed & implemented using cisco packet tracer, and communication between different subnets was verified through a router.